

SEOUL AB, Korea, Republic of

WMO: 471110

Lat: 37.446N

Lon: 127.114E

Elev: 92

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	13.6	-7.4	3.7	16.5	-3.9	4.5	19.2	13.7	30.8	12.2	29.2	4.4	330	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	92.8	78.5	89.8	76.9	87.6	75.6	80.1	88.0	79.1	86.0	78.1	84.4	6.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	148.8	83.5	77.2	141.8	82.6	75.8	135.3	81.5	43.9	89.0	42.7	85.6	41.6	84.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	11.8	10.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	27.9	32.4	43.1	53.8	64.1	72.8	78.5	79.8	70.1	57.9	45.8	31.2	
	DBStd	18.96	7.28	7.91	6.45	6.68	5.00	4.21	4.37	4.52	5.11	5.98	7.47	8.12	
	HDD50	2208	687	493	233	37	0	0	0	0	0	11	164	583	
	HDD65	5032	1152	912	680	338	78	4	0	0	14	230	576	1048	
	CDD50	3998	0	2	18	152	438	684	885	923	603	255	38	0	
	CDD65	1345	0	0	0	3	51	237	420	458	167	9	0	0	
	CDH74	11413	0	0	1	51	498	1719	3673	4376	1040	55	0	0	
	CDH80	3809	0	0	0	4	92	488	1297	1711	214	3	0	0	
Wind	WSAvg	4.2	3.9	4.3	5.2	5.1	4.6	3.9	4.4	4.1	3.5	3.5	3.8	4.1	
Precipitation	PrecAvg	54.0	0.7	1.1	1.7	3.2	3.7	5.7	15.2	11.4	6.6	1.9	1.7	1.0	
	PrecMax	84.2	2.5	3.1	5.1	7.4	9.4	17.0	34.5	34.6	23.1	7.3	5.0	2.3	
	PrecMin	31.6	0.0	0.0	0.2	0.8	0.5	1.1	5.0	2.4	0.8	0.0	0.3	0.2	
	PrecStd	12.3	0.6	0.8	1.1	1.7	1.8	3.7	6.9	7.5	5.8	1.5	1.0	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	58.8	68.7	79.1	86.1	91.0	95.4	96.5	87.7	79.1	68.4	53.9	
		MCWB	41.0	52.9	53.9	59.6	65.9	71.3	77.6	79.3	74.2	67.3	57.2	46.6	
	2%	DB	44.7	53.4	62.8	75.1	82.5	87.7	91.8	93.3	84.6	75.1	64.2	50.0	
		MCWB	37.8	44.8	50.3	59.3	63.7	70.6	77.9	79.5	71.7	63.7	55.2	44.0	
	5%	DB	42.5	48.6	58.8	71.5	79.2	85.3	89.5	91.1	82.5	71.7	60.9	46.6	
		MCWB	36.1	41.4	47.4	55.6	62.5	69.5	77.1	78.5	70.5	61.1	52.8	41.8	
	10%	DB	39.3	44.7	55.1	67.9	76.7	82.6	86.4	88.2	80.3	69.5	57.5	44.2	
		MCWB	34.0	38.1	45.1	53.8	61.5	68.7	76.1	77.3	69.4	59.8	51.1	39.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	55.2	57.0	64.9	70.7	75.6	81.2	82.2	78.1	69.5	60.9	49.3
			MCDB	45.8	56.8	63.9	73.8	80.1	83.8	89.8	90.4	83.8	75.1	65.7	51.2
		2%	WB	39.2	45.9	52.9	60.9	67.2	73.7	79.7	80.7	75.6	66.6	57.0	45.3
			MCDB	43.0	50.2	60.3	71.7	76.5	82.1	87.2	89.1	80.7	71.8	62.1	48.4
5%		WB	36.7	41.8	49.2	57.9	65.2	72.4	78.7	79.8	73.7	63.6	54.7	42.1	
		MCDB	41.0	47.2	56.8	67.6	74.7	80.4	85.3	87.3	79.0	68.9	59.4	45.9	
10%		WB	34.3	39.0	45.8	55.5	63.3	71.0	77.7	78.8	71.9	61.1	52.1	39.3	
		MCDB	38.6	44.2	53.2	64.7	73.0	78.7	84.0	85.5	77.3	67.3	56.8	42.8	
Mean Daily Temperature Range	5% DB	MDBR	16.8	16.9	18.9	21.1	21.0	17.8	13.4	14.1	17.3	19.7	17.1	15.4	
		MCDBR	18.5	21.6	24.8	27.0	26.1	22.0	17.7	17.2	19.0	21.5	20.6	17.6	
		MCWBR	12.7	13.8	14.1	13.5	12.0	8.9	7.1	6.7	8.2	11.3	13.0	13.2	
	5% WB	MCDBR	16.3	18.3	21.3	22.1	20.1	16.9	13.7	14.4	14.6	17.6	17.9	16.0	
		MCWBR	12.7	13.6	13.9	12.8	11.3	7.9	6.4	6.4	7.4	10.6	13.1	13.3	
Clear-Sky Solar Irradiance	taub	0.444	0.498	0.584	0.628	0.640	0.712	0.678	0.591	0.516	0.508	0.489	0.440		
	taud	1.931	1.837	1.694	1.647	1.669	1.592	1.708	1.905	2.034	1.937	1.927	1.962		
	Ebn at Noon	218	223	217	217	218	203	208	224	234	218	202	208		
	Edh at Noon	45	56	71	78	78	84	75	60	50	50	45	41		
All-Sky Solar Radiation	RadAvg	790	1034	1346	1569	1766	1648	1284	1366	1300	1133	775	696		
	RadStd	49	83	136	113	161	164	194	194	144	81	76	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+90

Nomenclature: See separate page